



**Faculty of Engineering & Technology Electrical & Computer Engineering
Department**

**Circuits and Electronics Laboratory - ENEE2103
PreLab Exp2: Circuit Laws and Theorems.**

Prepared By:

Aya Dahbour 1201738

Instructor: Dr. Nasser Ismail.

Assistant: Eng. Bilal Ismail

Section: 3

Date: 1/11/2022

A. KVL,KCL

Consider $R_x=R_1=R_4=1k\Omega$, $R_5=3.3k\Omega$, $R_6=4.7k\Omega$ and $V_s= 15V$

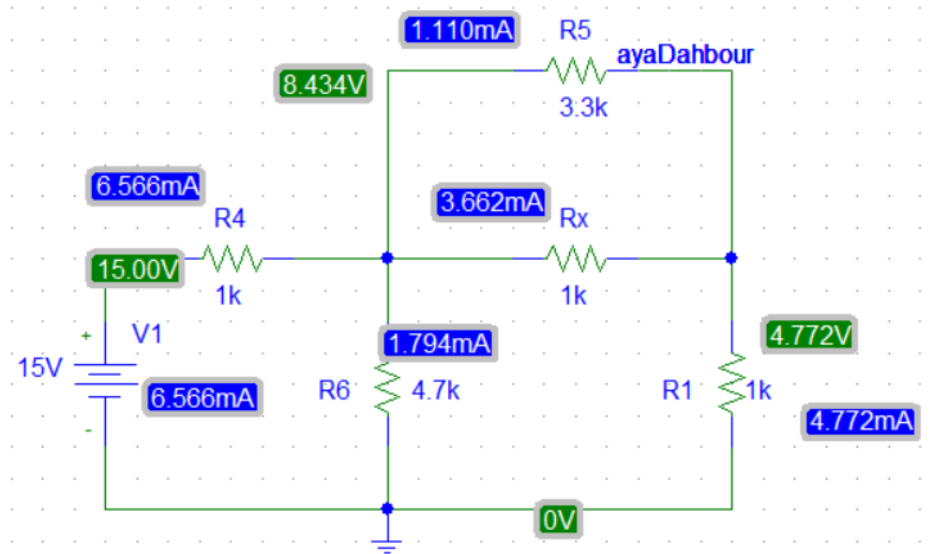


Figure 1 : KVL,KCL circuit

$$V1 = 4.772v$$

$$V4 = 15 - 8.434v = 6.57v$$

$$V5 = V_x = 8.434v - 4.772v = 3.66v$$

$$V6 = 8.434v$$

V_s		R1		R4		R5		R6		Rx	
		V1	I1	V4	I4	V5	I5	V6	I6	Vx	Ix
15V	$R_x=1k$	4.772	4.772	6.57	6.566	3.66	1.11	8.434	1.794	3.66	3.662

KVL,KCL table

B. Voltage & Current Division :

I. Current division

replace R1 by a short circuit and set R_x to 1k.

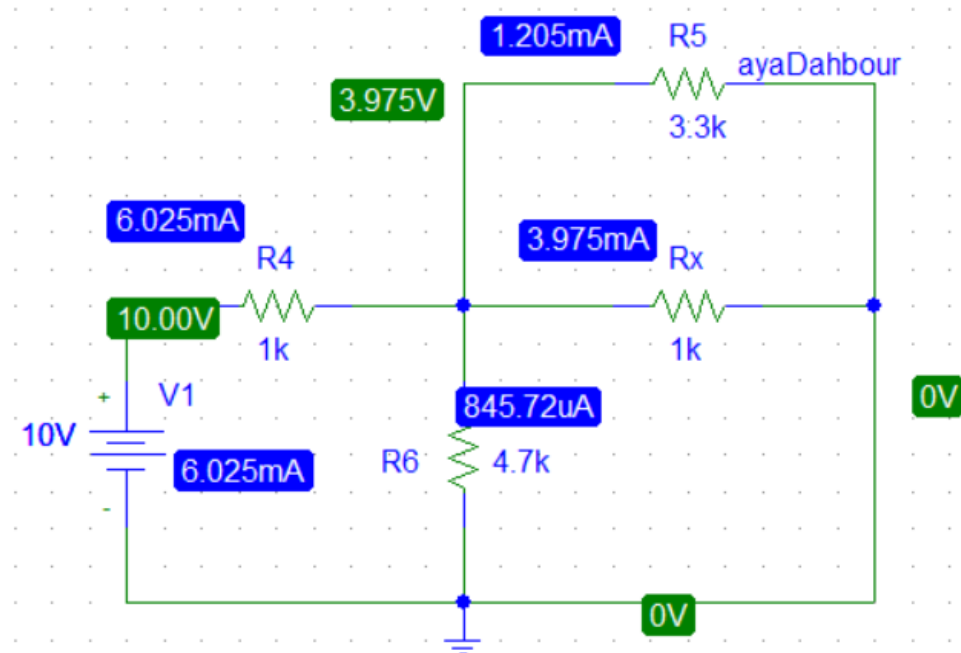


Figure 2: Voltage & Current Division circuit

$$V_4 = 10\text{v} - 3.975\text{v} = 6.025\text{v}$$

$$V_5 = V_x = V_6 = 3.975\text{v}$$

Vs (volt)	Pot.	I4	I5	I6	Ix
10	Rx	6.025mA	1.205mA	845.72uA	3.975mA

Voltage & Current Division table

C. Superposition :

First with two sources :

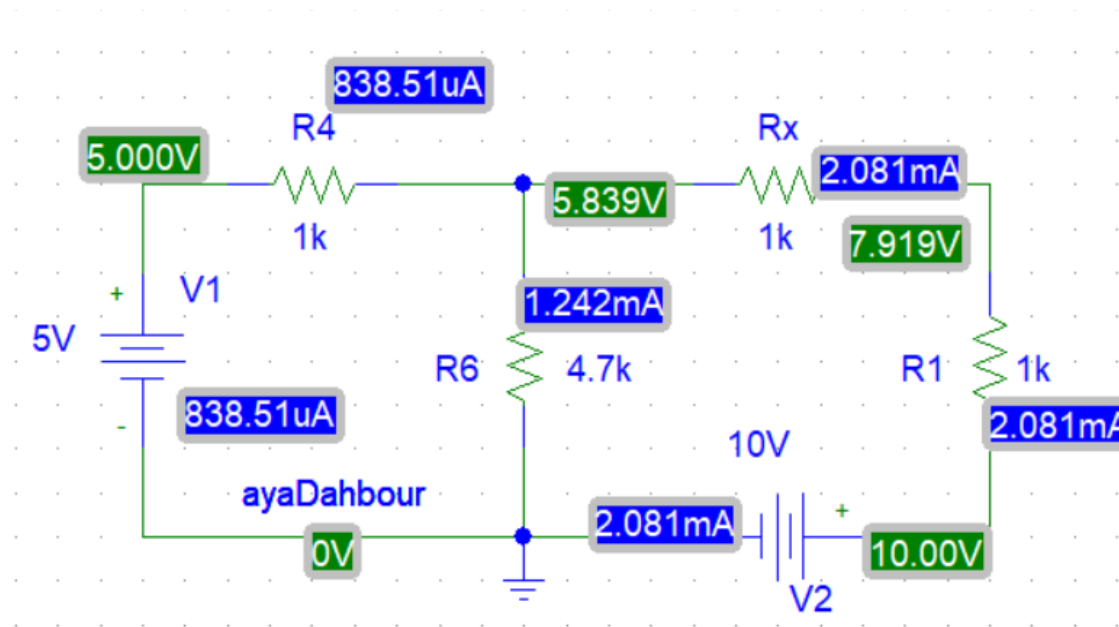


Figure 3: Superposition circuit1

Then kill V1 :

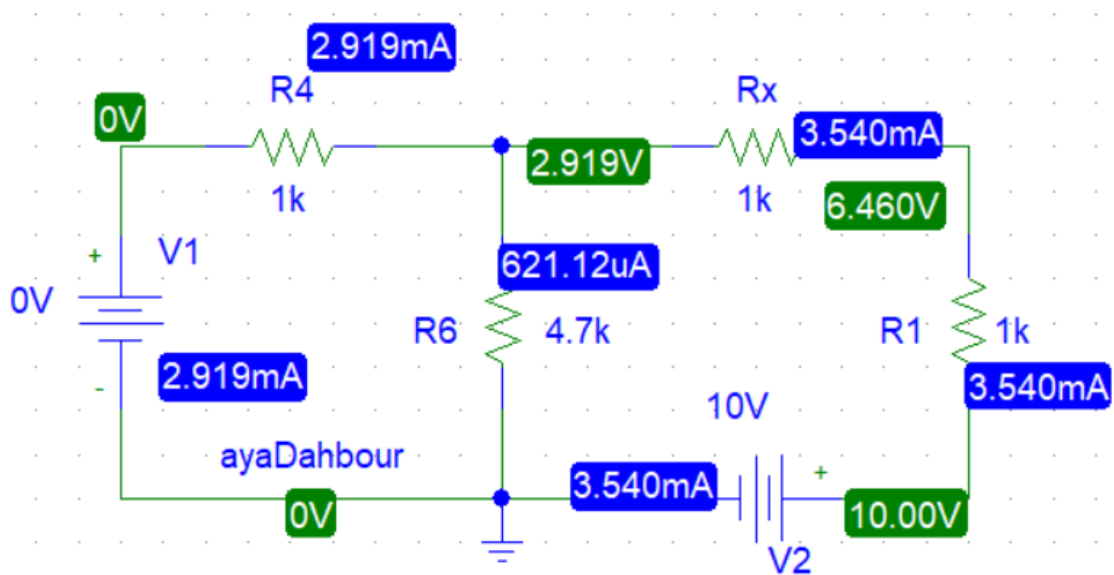


Figure 4: Superposition circuit2

Finally kill V2 :

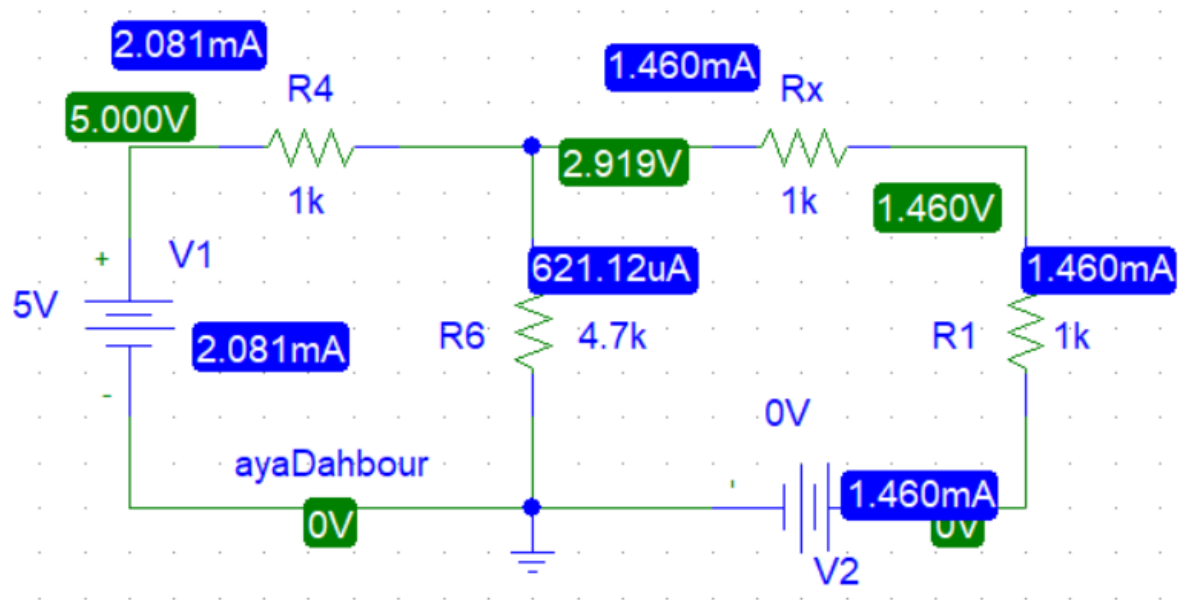


Figure 5:Superposition circuit3

Vs1(volt)	Vs2 (volt)	V6 (volt)	I6 (mA)
5	10	5.938v	1.242mA
0	10	2.919v	621.12uA
5	0	2.919v	621.12uA

Superposition table

D. Thevinin and Norton equivalent circuits:

As shown below the voltage across R1 = $10 - 7.919 = 2.081\text{v}$.

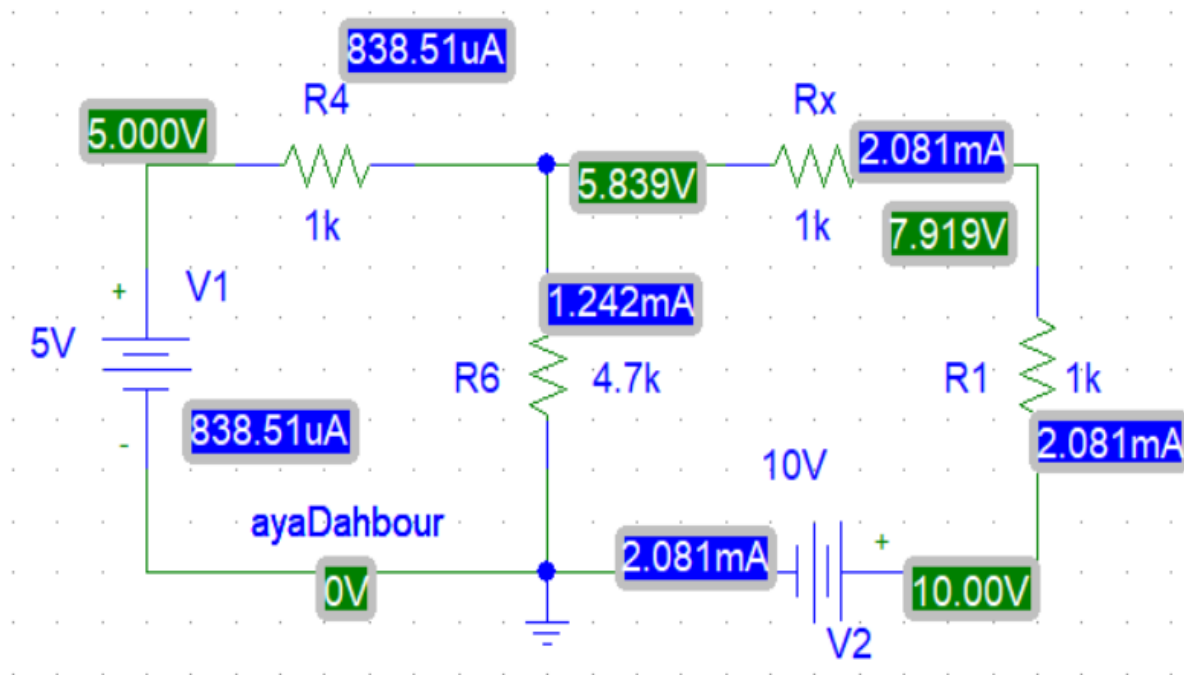


Figure 6: circuit before thevinin

To get open circuit in ps spice resistor that has a high value (150 tera) and find Vth :

$$V_{th} = 10 - 4.123 = 5.877\text{v}$$

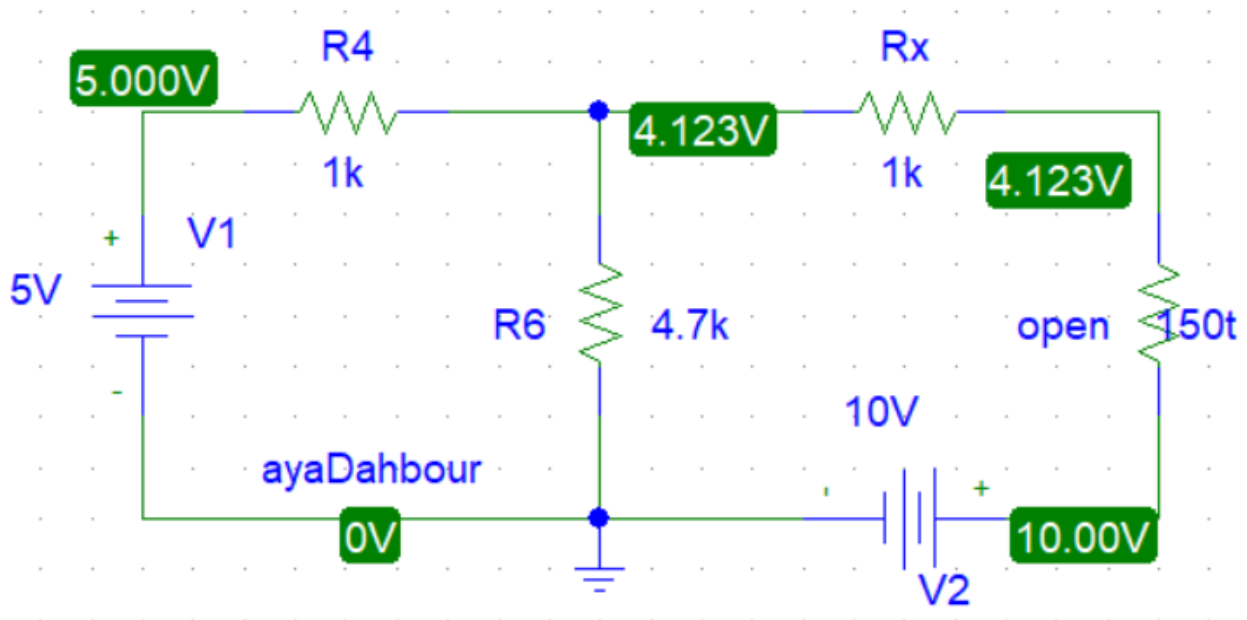


Figure 7: Vth

Then find R_{th} :

$$R_{th} = (R_6 \parallel R_4) + R_x$$

$$= (1K + 4.7K) + 1K = 1.825K \Omega$$

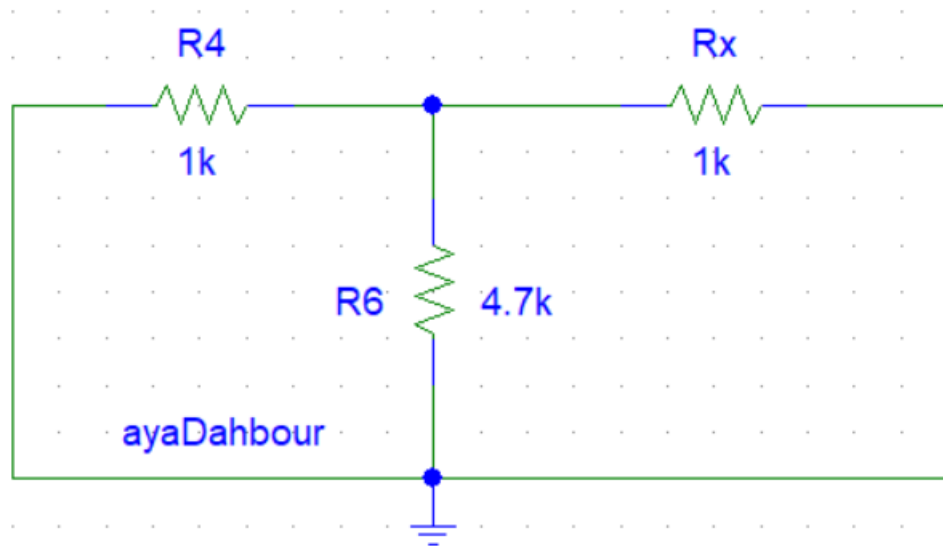


Figure 8: R_{th}

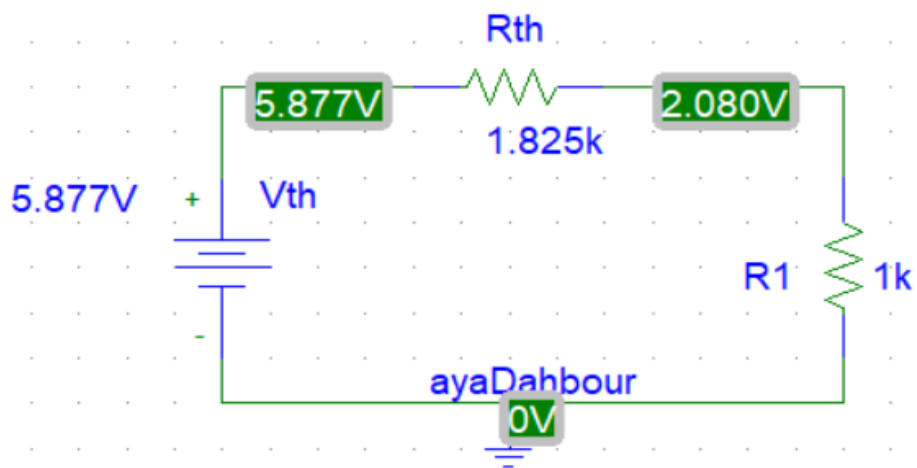


Figure 9: thevinin circuit

It can be deduced that the circuit has an equivalent circuit with one voltage source and one resistor that produces the same outputs as the original since the voltage on resistor R_1 is equal to $2.08V$.